DATA ANALYTICS PORTFOLIO REPORT

Healthcare Operations & Revenue Analytics

SQL Server + Power BI | Patient Volume · Revenue · Length of Stay · Hospital Performance

Hafiz Arslan Shafique hafizarslan3195@gmail.com linkedin.com/in/hafiz-arslan-shafique-bc240203664 github.com/Hafiz-Arslan-Shafique

Executive Summary

This project combines SQL Server data preparation with a Power BI dashboard to examine patient volume, healthcare revenue, hospital performance, medical-condition revenue, demographics, and length of stay across a 55,501-row healthcare dataset (2019–2024). The SQL workflow validates the raw records, standardizes patient names, calculates length of stay, detects and removes duplicate visits, corrects negative billing values, and produces business-oriented revenue aggregations. Power BI then presents the resulting analysis through KPI cards, a hospital revenue ranking, a medical-condition revenue treemap, demographic breakdowns, and a length-of-stay target gauge.

> **Business Problem:** Healthcare operations teams need to monitor patient volume, billing performance, hospital revenue, medical-condition revenue, and length of stay in a single, trustworthy analytical view — so operational and financial patterns can be reviewed with confidence rather than against raw, duplicated, or error-filled records.

Project Objectives

- Validate and clean healthcare records in SQL Server.
- Create a deduplicated analytical table suitable for reporting.
- Analyze revenue and patient volume by medical condition, insurer, admission type, and hospital.
- Present the key operational metrics through an interactive Power BI dashboard.

Dataset Overview

Item	Supplied Detail
Rows	55,501
Date range	2019–2024
Admission types	Emergency, Urgent, Elective
Test results	Normal, Abnormal, Inconclusive
Core fields	Patient demographics, medical condition, admission/discharge dates, doctor, hospital, insurer, billing amount, room number, medication, test results

SQL Analysis

The SQL workflow creates **HealthcareDB** and a raw import schema, verifies the imported data, builds **Clean_Healthcare** with standardized names and calculated length of stay, identifies duplicate visits, builds **Final_Healthcare** using **ROW_NUMBER()**-based deduplication, validates categories and NULLs, checks numerical ranges, corrects negative billing, and produces the analytical summaries that feed the dashboard.

EDA & Data Cleaning

EDA includes `TOP 5` inspection, `COUNT(*)` checks, `DISTINCT` category checks, duplicate detection, NULL checks, and MIN/MAX/AVG range checks. Cleaning includes `UPPER(TRIM(Name))` standardization, `DATEDIFF`-based length-of-stay calculation, exclusion of missing dates and negative LOS, `ROW_NUMBER()`-based deduplication, and `ABS()`-based correction of negative billing values.

Important SQL Techniques

The strongest techniques demonstrated are **CTEs**, **ROW_NUMBER()**, **PARTITION BY**, `GROUP BY`, `HAVING`, `CASE`-based conditional aggregation, `DATEDIFF`, `CAST`, `TOP`, `ORDER BY`, `UPDATE`, and `ABS`. No views, joins, `RANK()/DENSE_RANK()`, running totals, or `SUM() OVER()` are present in the supplied workflow — the project intentionally documents only what was actually built, rather than overstating scope.

Power BI Development

The supplied PBIX uses **Final_Healthcare** as the visible analytical table in the model layout. The dashboard contains KPI cards, a top-hospital revenue bar chart, a medical-condition revenue treemap, a patient-by-gender donut, a patient-by-condition stacked bar chart, a length-of-stay-vs-target gauge, and slicers for gender, visit/admission type, and medical issue/condition.

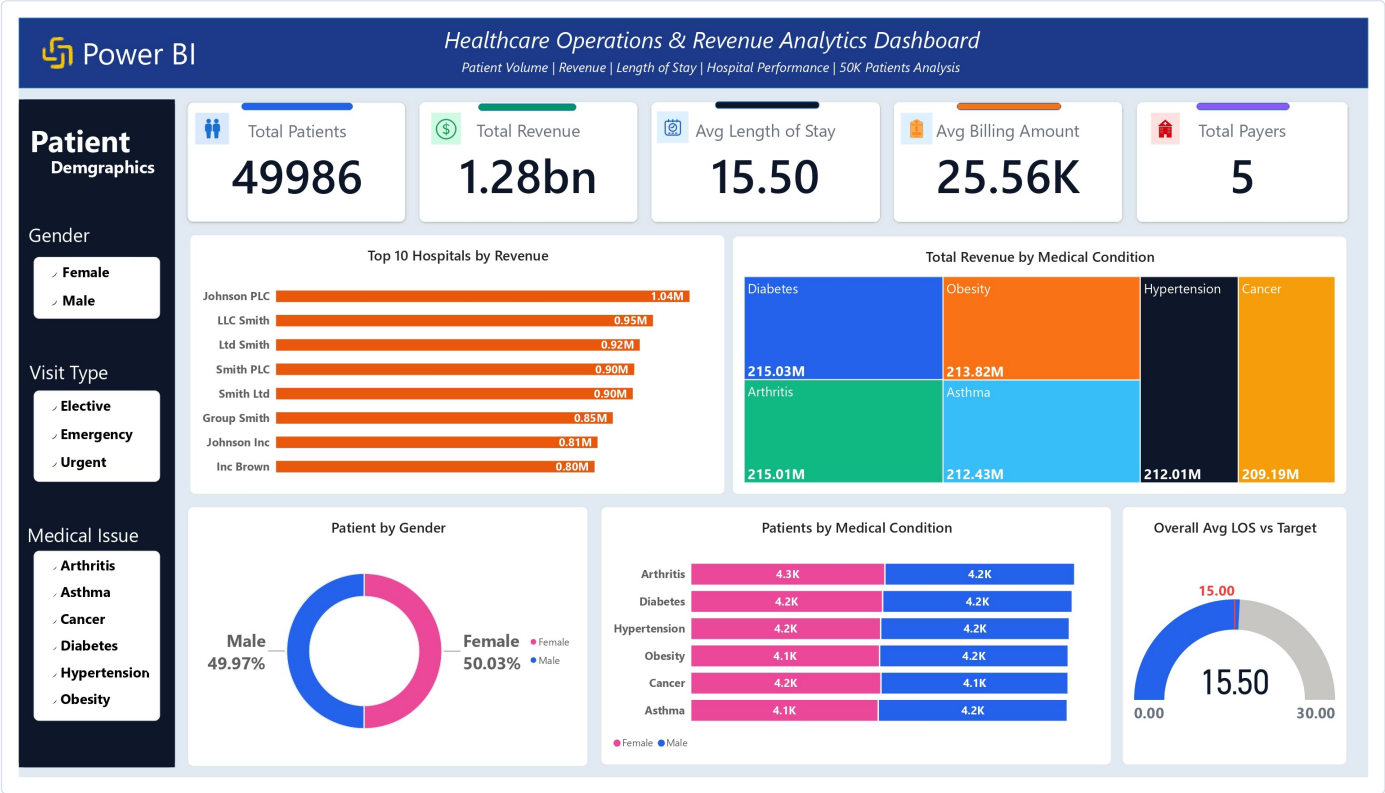
Dashboard KPIs

49,986	1.28bn	15.50	25.56K	5	15.00
TOTAL PATIENTS	TOTAL REVENUE	AVG LENGTH OF STAY	AVG BILLING AMOUNT	TOTAL PAYERS	LOS TARGET

Key Business Insights

- **Diabetes carries the highest visible condition revenue** at 215.03M, while Cancer sits lowest at 209.19M — a relatively narrow spread, showing revenue is distributed fairly evenly rather than concentrated in one condition.
- **Gender distribution is almost perfectly balanced** — Female at 50.03% vs. Male at 49.97%.
- **Patient volume is tightly clustered across conditions**, with each visible condition/gender segment sitting in the 4.1K–4.3K range — no condition dominates caseload.
- **Johnson PLC leads hospital revenue** at 1.04M, followed by LLC Smith at 0.95M.
- **Average length of stay runs above target** — 15.50 vs. a 15.00-day target, a 0.50-day gap.

Dashboard Screenshot



Source: supplied Power BI dashboard image. All KPI values and visible observations in this report are reproduced only from the supplied image and dataset — nothing is estimated or fabricated.

Business Recommendations

- **Investigate the 0.50-day length-of-stay gap** against the 15.00-day target, and identify opportunities to improve discharge throughput.
- **Benchmark leading vs. lower-ranked hospitals** using patient volume and billing patterns to isolate what top performers like Johnson PLC are doing differently.
- **Use condition-level LOS and average billing alongside total revenue** when prioritizing operational improvement areas, rather than revenue totals alone.
- **Retain the duplicate-detection and negative-billing checks** as standing, recurring data-quality controls rather than one-time cleanup steps.

Results

The project converts a 55,501-row raw healthcare dataset into a cleaned, deduplicated analytical layer (**Final_Healthcare**) and a focused management dashboard. It demonstrates practical SQL Server data-quality and analytical skills — validation, standardization, deduplication, and business-facing aggregation — together with Power BI dashboard design and visualization.

Conclusion

This portfolio project demonstrates an end-to-end healthcare analytics workflow without claiming unsupported SQL or Power BI functionality. The **README** provides the recruiter-facing project summary, while this report provides deeper documentation, methodology detail, and the dashboard visual evidence behind it.

Hafiz Arslan Shafique

Data Analyst | SQL Server · Power BI · Excel

Email: hafizarslan3195@gmail.com LinkedIn: Profile GitHub: Profile Phone: +966 57 959 4038